

Bitwise linear trail through **WARP** found by SAT

CiVerLy

July 21, 2026

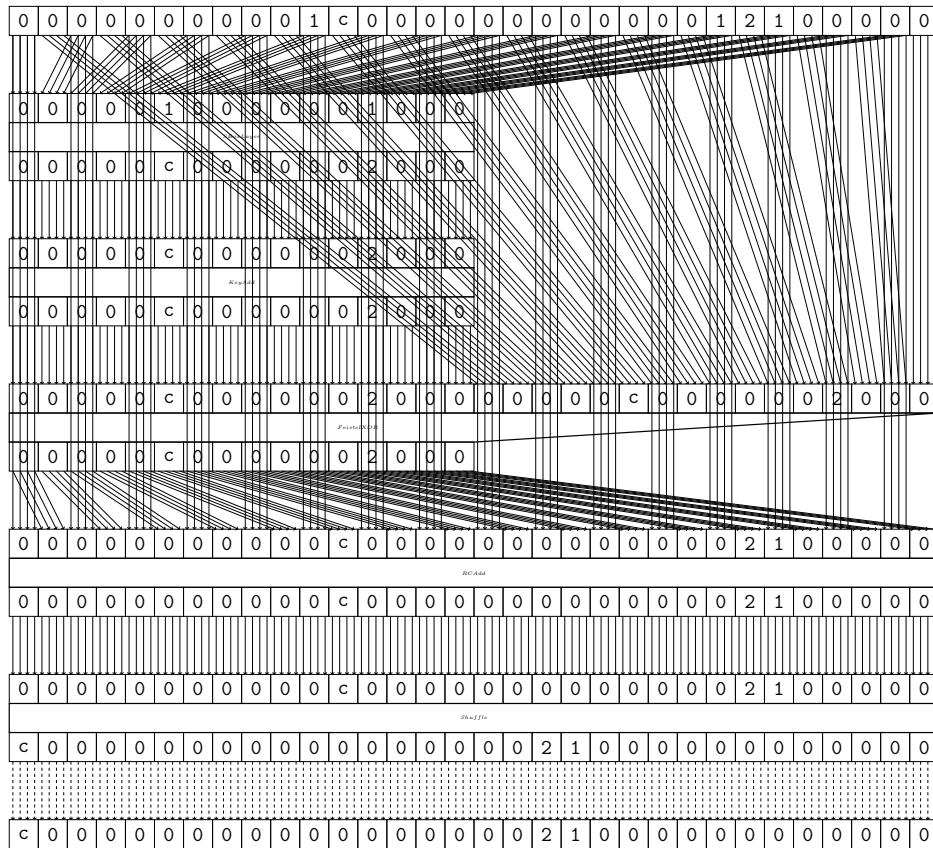
1 WARP

Maximal linear correlation: 2^{-8}

0	0	0	0	0	0	0	0	0	0	0	1	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	1	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	1	0	0	0	0	0
unpermut																																		
c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	c	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	c	
unpermut																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	
unpermut																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																		
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	
unpermut																																		
0	0	0	0	1	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
0	0	0	0	1	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
unpermut																																		
0	1	0	0	0	0	0	0	c	0	0	d	0	0	0	0	0	0	2	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																																		
0	1	0	0	0	0	0	0	c	0	0	d	0	0	0	0	0	0	2	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	

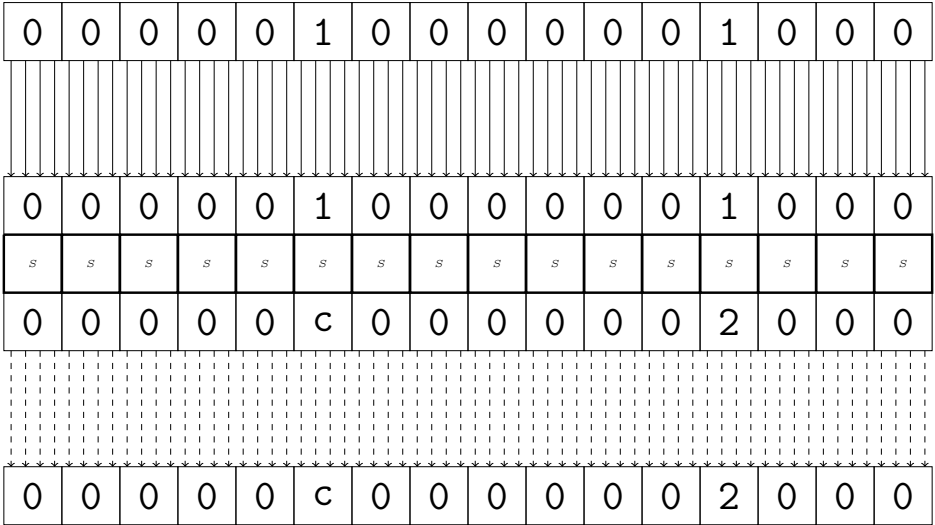
2 warp_round

Maximal linear correlation: $2^{-2.0}$



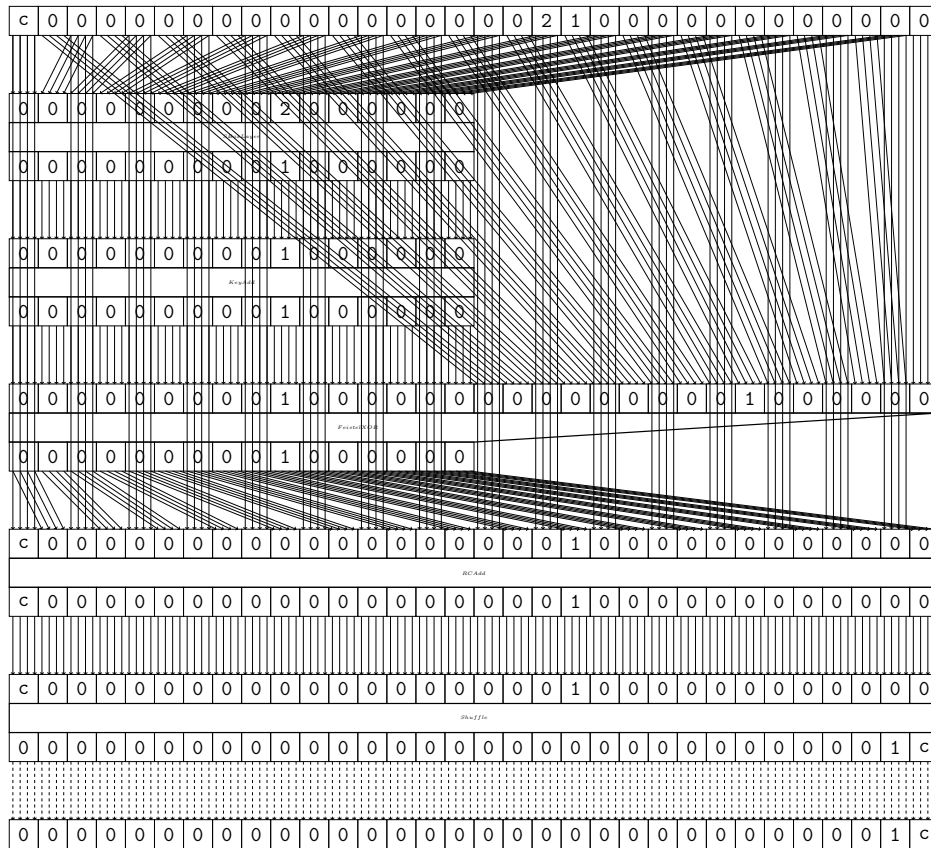
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



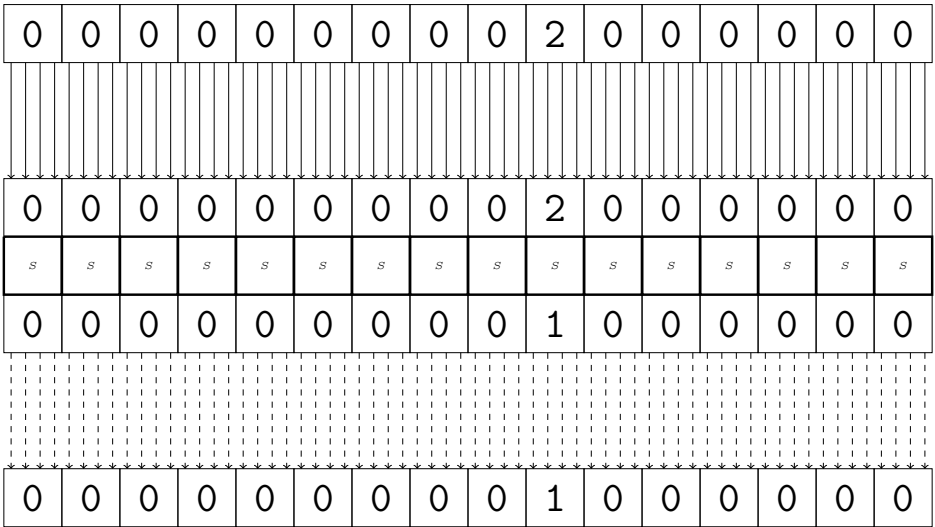
4 warp_round

Maximal linear correlation: $2^{-1.0}$



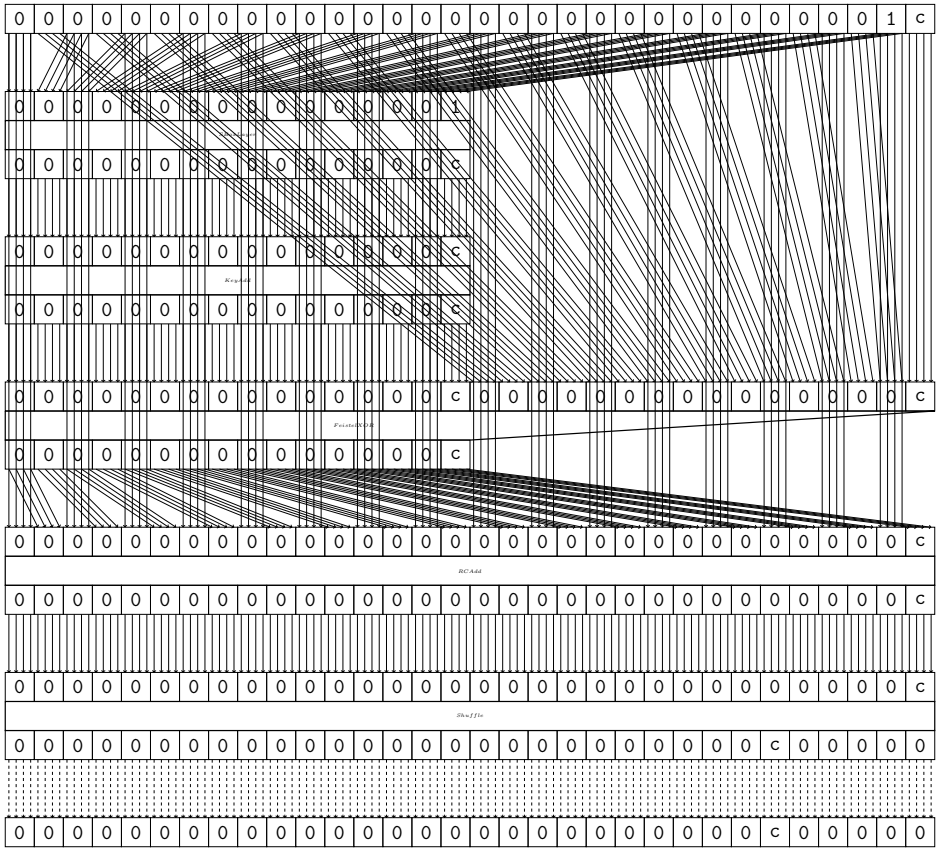
5 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



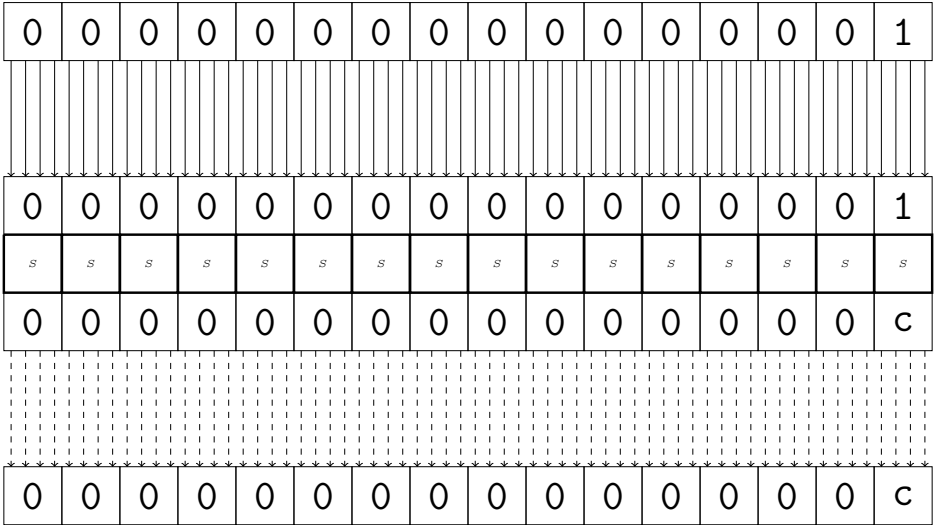
6 warp_round

Maximal linear correlation: $2^{-1.0}$



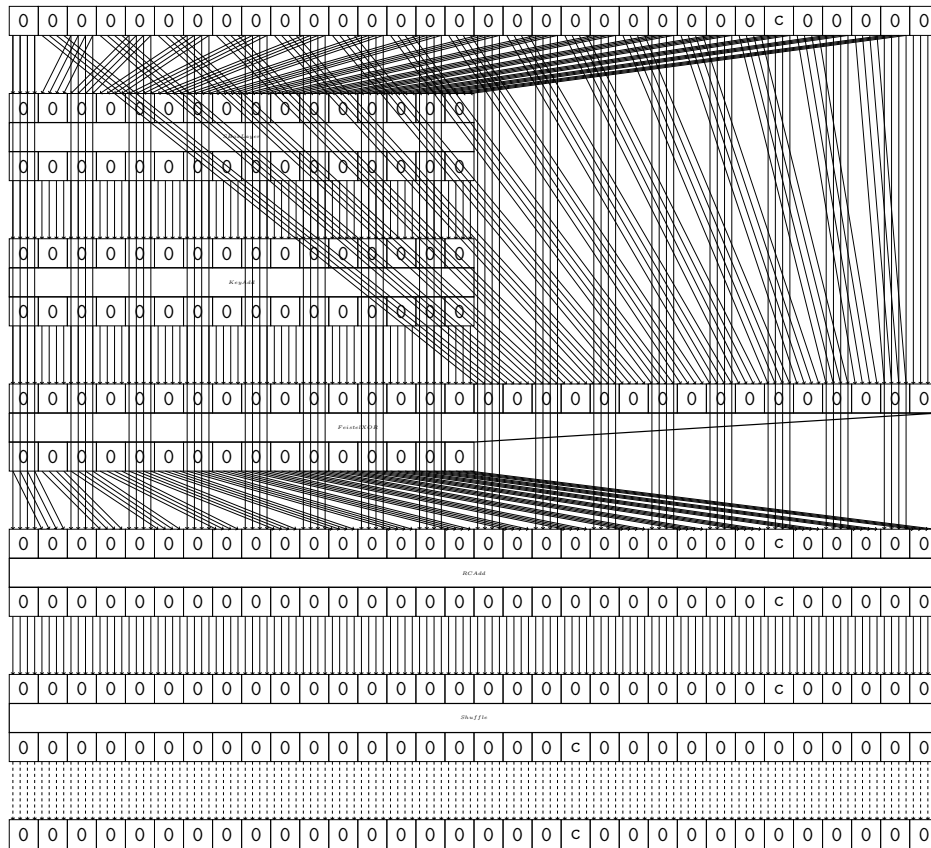
7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



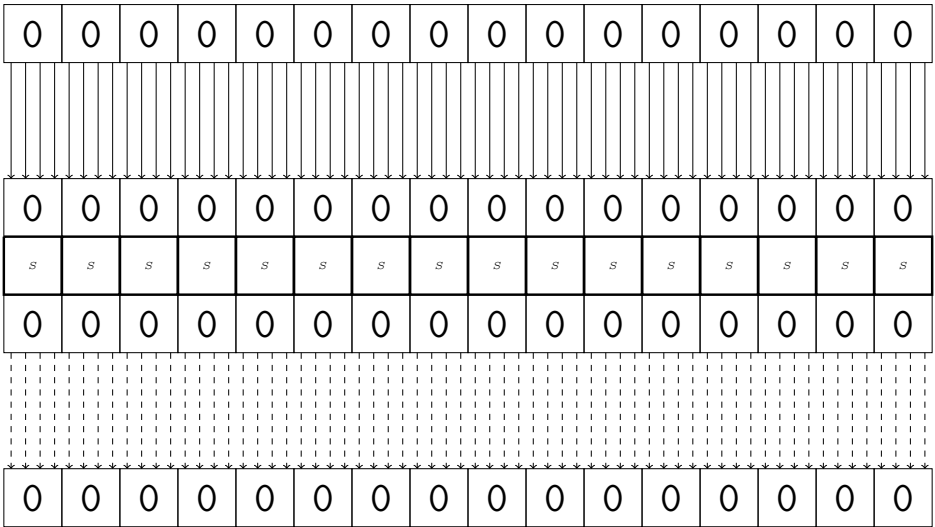
8 warp_round

Maximal linear correlation: $2^{-0.0}$

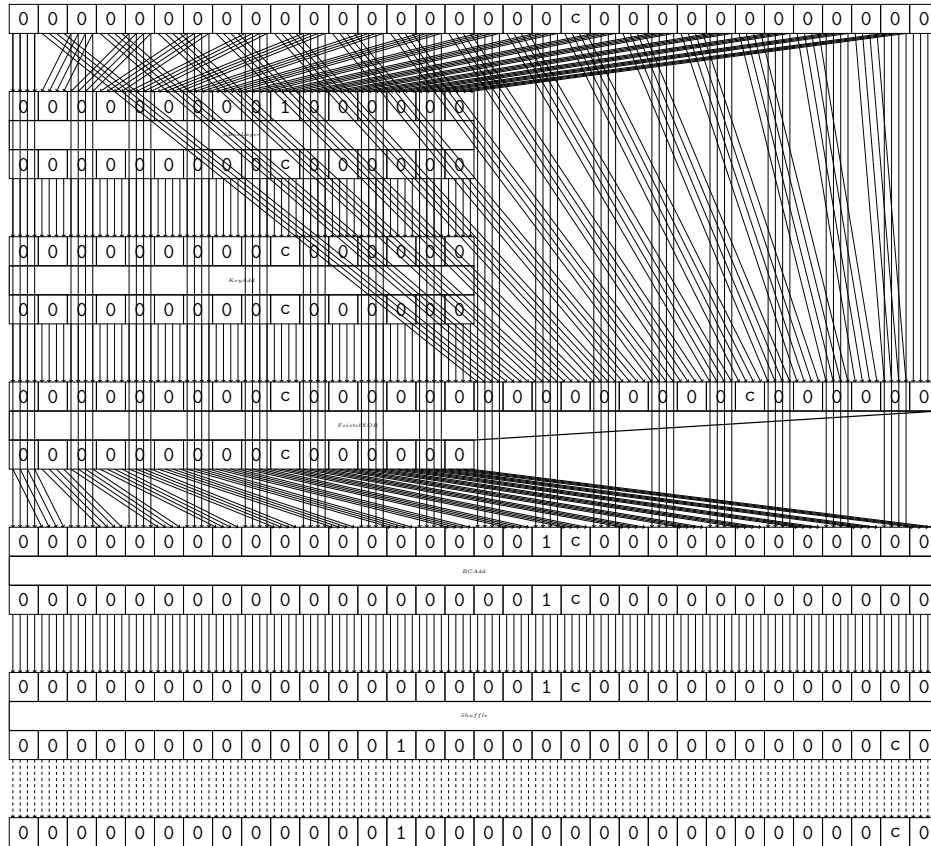


9 SBoxLayer

Maximal linear correlation: $2^{-0.0}$

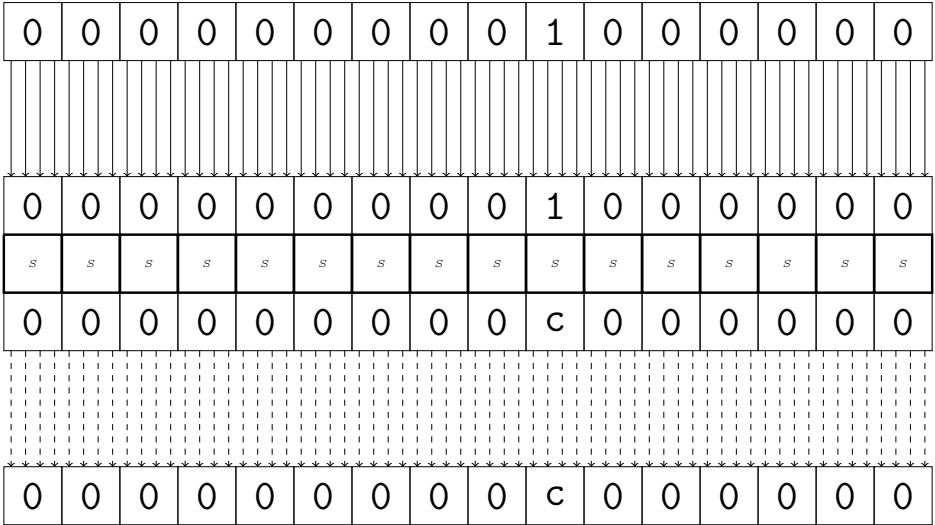


10 warp_round

Maximal linear correlation: $2^{-1.0}$ 

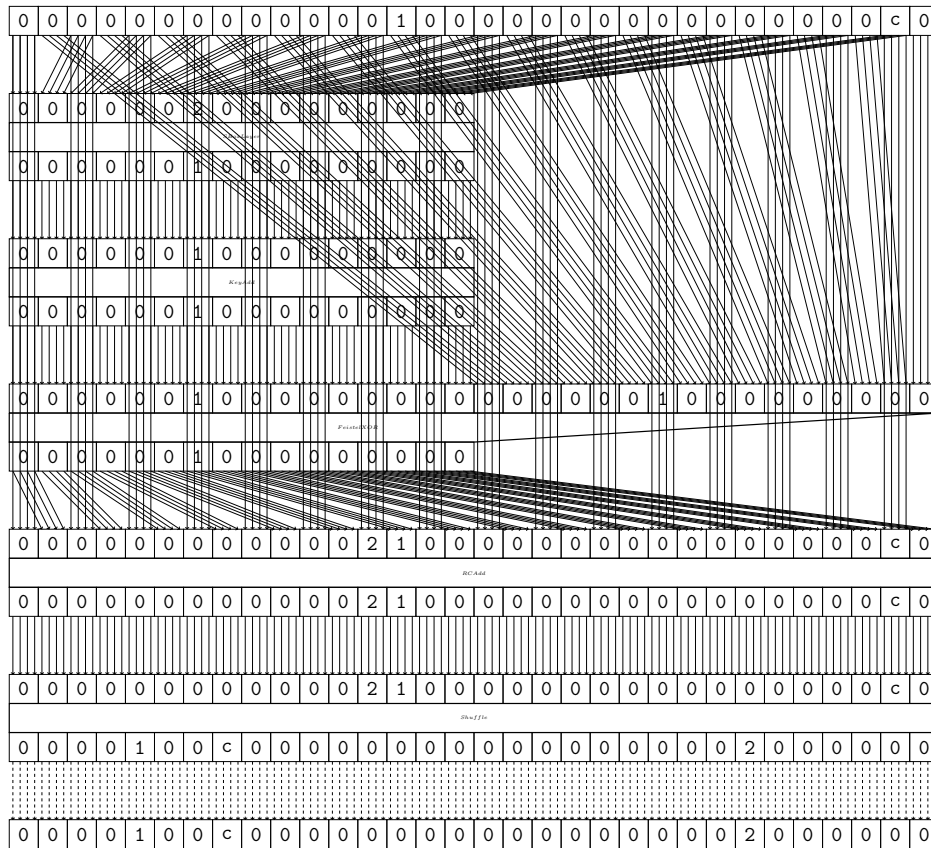
11 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



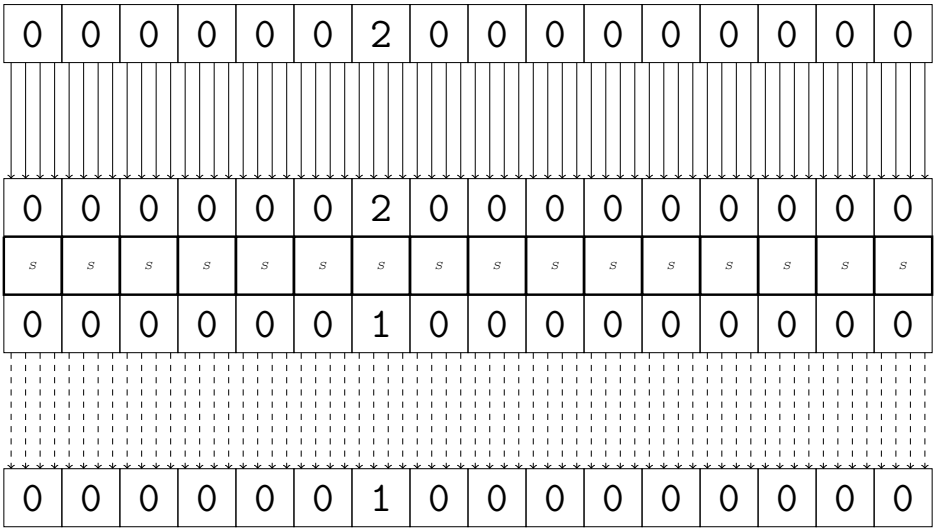
12 warp_round

Maximal linear correlation: $2^{-1.0}$



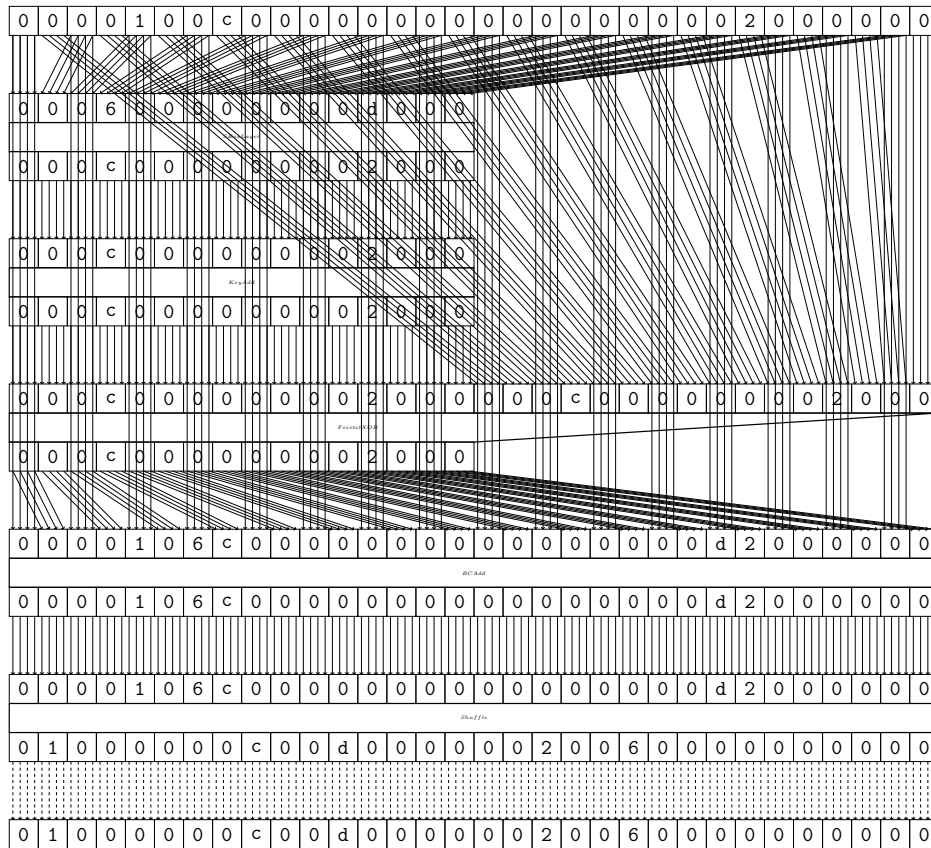
13 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



14 warp_round

Maximal linear correlation: $2^{-2.0}$



15 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

